Desert Eagle / Magnum Research

Magnum Research / IWI • Mark XIX Magnum Handguns (.50 AE, .44 Magnum, .357 Magnum)

Manufacturer:	Magnum Research / IWI	**Category:**	Pistols
Action Type:	Gas-Operated Rotating Bolt Magnum Pistol	**Era / Service:**	Modern Heavy Magnum (1983–Present)
Calibers:	.50 AE, .44 Magnum, .357 Magnum	**Covered Models:**	Desert Eagle Mark XIX .50 AE, Desert Eagle .44 Magnum, Desert Eagle .357 Magnum

KEY ARMORER TASKS & BENCH OPERATIONS

Core Procedures: Gas Piston Chamber Cleanout • Rotating 3-Lug Bolt Maintenance • Dual Recoil Spring Rod Assembly • Extractor Spring Tuning • Gas Cylinder Scraping

CRITICAL ARMORER TOLERANCES & SPECIFICATIONS

SPECIFICATION / CLEARANCE	ARMORER TOLERANCE & BENCH SPECIFICATION
OPERATING SYSTEM	Gas-operated short stroke piston under barrel with 3-lug rotating bolt (rifle-style lockup)
GAS SYSTEM MAINTENANCE	Gas port located just forward of chamber; lead cast bullets STRICTLY PROHIBITED (lead vapors will permanently plug gas port)
DUAL RECOIL SPRINGS	Twin captive spring guide rods on either side of barrel lug
BOLT LOCKING	Three heavy locking lugs lock into barrel extension identical to an AR-15 / rifle design

IMMEDIATE ACTION & MALFUNCTION DIAGNOSTIC MATRIX

MALFUNCTION CONDITION	PROBABLE CAUSES	ARMORER BENCH REMEDY
Stoppage 1: Short Recoil / Failure to Eject (Lead Clogged Gas Port) **Pistol cycles partially rearward, spent case remains in ejection port, bolt fails to reach rear recoil stop**	• Shooting non-jacketed lead bullets; molten lead vapors clogged the micro gas port under the barrel. • Limp-wristing heavy recoil absorption.	**Gunsmith Action: Check gas piston for free travel. Remove barrel. Soak gas port in lead solvent. Use drill shank or ultrasonic cavitation to clear gas passageway. Warn user: ONLY full metal jacket or copper-plated bullets permitted in gas-operated pistols.**

CURATED BENCH VIDEO REFERENCES

PROCEDURE TITLE	INSTRUCTOR / CREATOR	TECHNICAL PROCEDURE SUMMARY
Desert Eagle .50 AE Complete Disassembly, Cleaning & Gas System	Forgotten Weapons / Magnum Bench	Teardown of the rotating bolt, gas piston inspection, twin recoil rod replacement, and proper lubrication of the heavy steel slide tracks.